Task 2: LLM Logic Onboarding
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Objective (setup)

1. Joining the OpenAI namespace and getting an OpenAI key
2. Understand SWI-Prolog. Use an interpreter in Python (janus-swi)
3. Baseline experiment translating Natural Language to Symbolic Reasoning

Main Goal

Achieving a Natural Language to Symbolic Reasoning translator.
To determine its effectiveness, we ran an experiment where we created a file "nlp_prolog.py". This Python file takes a question, for example, "When is driving better than walking?" and returns symbolic reasoning based on generated Prolog code and a generated query.

Steps: user question -> LLM response -> LLM generates a prolog of its response and the best query in JSON format -> query run against prolog using Janus-swi and symbolic representation is printed.

Process

The main problem we ran into is the step where the LLM had to transfer its response to JSON data containing (file.pl, best query). This step proved to be difficult because the initial model (gpt-3.5-turbo) struggled to generate a correct JSON format and meaningful data. I had to carefully prompt the engineer for the exact response I wanted.

Experiment 1 (gpt-3.5-turbo):
After evaluating over 20 responses. I crafted the best code I could. Then I ran a test using "when is?" type questions. The model proved to have a 3/4 success rate at producing JSON data that can be interpreted using Janus-swi.

Experiment 2 (gpt-4.1-mini):
I decided to pivot after I didn't have reliable results. I also carefully prompt engineered the AI JSON response better than experiment 1 after more experimentation. This time, I was getting better results! I ran an experiment 2 using meaningful random questions. For example: "What are the reasons that water is essential for the human body?" This time, I got a 5/5 success rate and more meaningful data than experiment 1.

Conclusion

Now I have an effective Natural Language to Symbolic Reasoning translator. I used a small sample size when testing, so there might be some future tweaks and corrections to robustify it. We can work on having more effective queries and testing more types of questions.

Results

You can see the results in my research diary under task2: https://colab.research.google.com/drive/1xS9LHUfySsJRc6W1wrtZvdDpgTs3jW3h#scrollTo=0no3YlBGR7w4.

The diary does not include the actual “nlp_prolog.py” that’s only saved locally so far.